$$\frac{h\nu}{c} = \frac{m_0 v}{\sqrt{1-(v^2/c^2)}} \tag{1}$$

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} \tag{2}$$

$$E = \sqrt{m_0^2 c^4 + h^2 \nu^2} \tag{3}$$

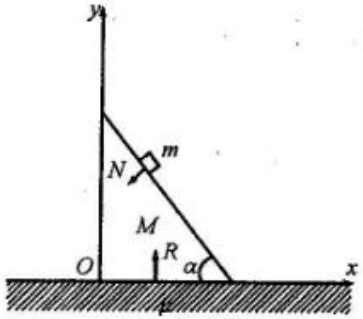
$$E_0 = h\nu + m_0 c^2 \tag{4}$$

$$E^2 - E_0^2 = (m_0^2 c^4 + h^2 \nu^2) - (h\nu + m_0 c^2)^2 = -2h\nu m_0 c^2 \neq 0 \tag{5}$$

$$h\nu \geqslant \left[\frac{m_0 c^2}{\sqrt{1-(v^2/c^2)}} - m_0 c^2\right] + W \tag{6}$$

$$\frac{1}{\sqrt{1-(v^2/c^2)}} \approx 1 + \frac{1}{2}\frac{v^2}{c^2} + \cdots \tag{7}$$

$$h\nu \geqslant \frac{1}{2} m_0 v^2 + W \tag{8}$$



$$ma_x = N\sin\alpha \tag{1}$$

$$ma_y = mg - N\cos\alpha \tag{2}$$

$$Ma_M = N\sin\alpha - \mu R \tag{3}$$

$$R = N\cos\alpha + Mg \tag{4}$$